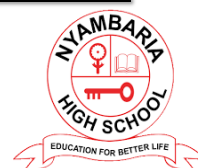
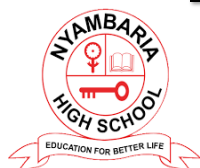


NYAMBARIA BOYS' 2025 PREDICTION



KCSE PROJECTION EXAM

PHYSICS

232/3

PAPER 3

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

INSTRUCTIONS TO CANDIDATES

1. Write your **name, index number, class, date** and **signature** in the spaces provided above.
2. This paper consists of two questions 1 and 2.
3. Answer all questions in the spaces provided.
4. Non-programmable calculators and mathematical tables may be used.
5. Show all your workings.

QUESTION 1	c	g	h	i	j	k	l
Maximum score	1	8	5	2	1	2	1
Candidates score							

QUESTION 2	b	c	d	e		GRAND TOTAL
Maximum score	3	2	6	9		
Candidates score						

QUESTION ONE

You are provided with the following

2 new dry cell size **D**

A cell holder

A switch

A millimeter of range **0 – 1mA**

A capacitor labelled **C**

8 connecting wires at least four with crocodile clips on one end

A stop watch

A carbon resistor labelled **R**

Proceed as follows

a) **Connecting** the circuit as shown in the **figure 1** below, where **P** and **Q** are crocodile clips

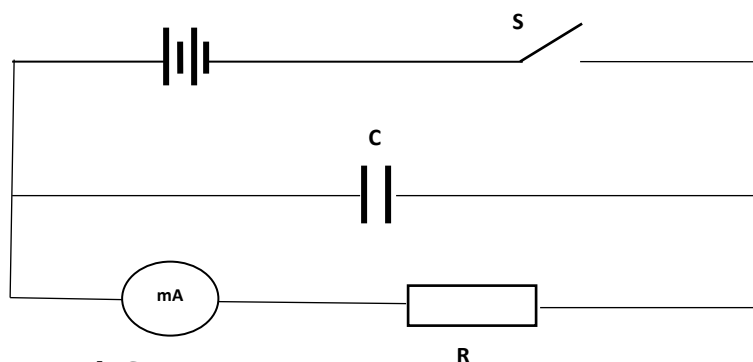


Figure 1

b) Close the switch **S**

c) Record the highest reading of the milliammeter **I₀** and then open the switch.

I₀ =.....

(1 mark)

d) Use **I₀** above to calculate $\frac{4}{5}I_0$, $\frac{3}{4}I_0$, $\frac{2}{3}I_0$, $\frac{1}{2}I_0$, $\frac{2}{5}I_0$, $\frac{1}{3}I_0$ and $\frac{1}{4}I_0$. Record in the **table 1** below.

e) Close switch **S** for a second time and observe the deflection in the milliammeter (*the pointer should rise back to the same initial value I₀.*)

f) Open switch **S** and at the same time start the stop watch to measure the time taken for the current to decrease to four fifth the value of **I₀**. i.e. $\frac{4}{5}I_0$. Record your value in the **table 1** below.

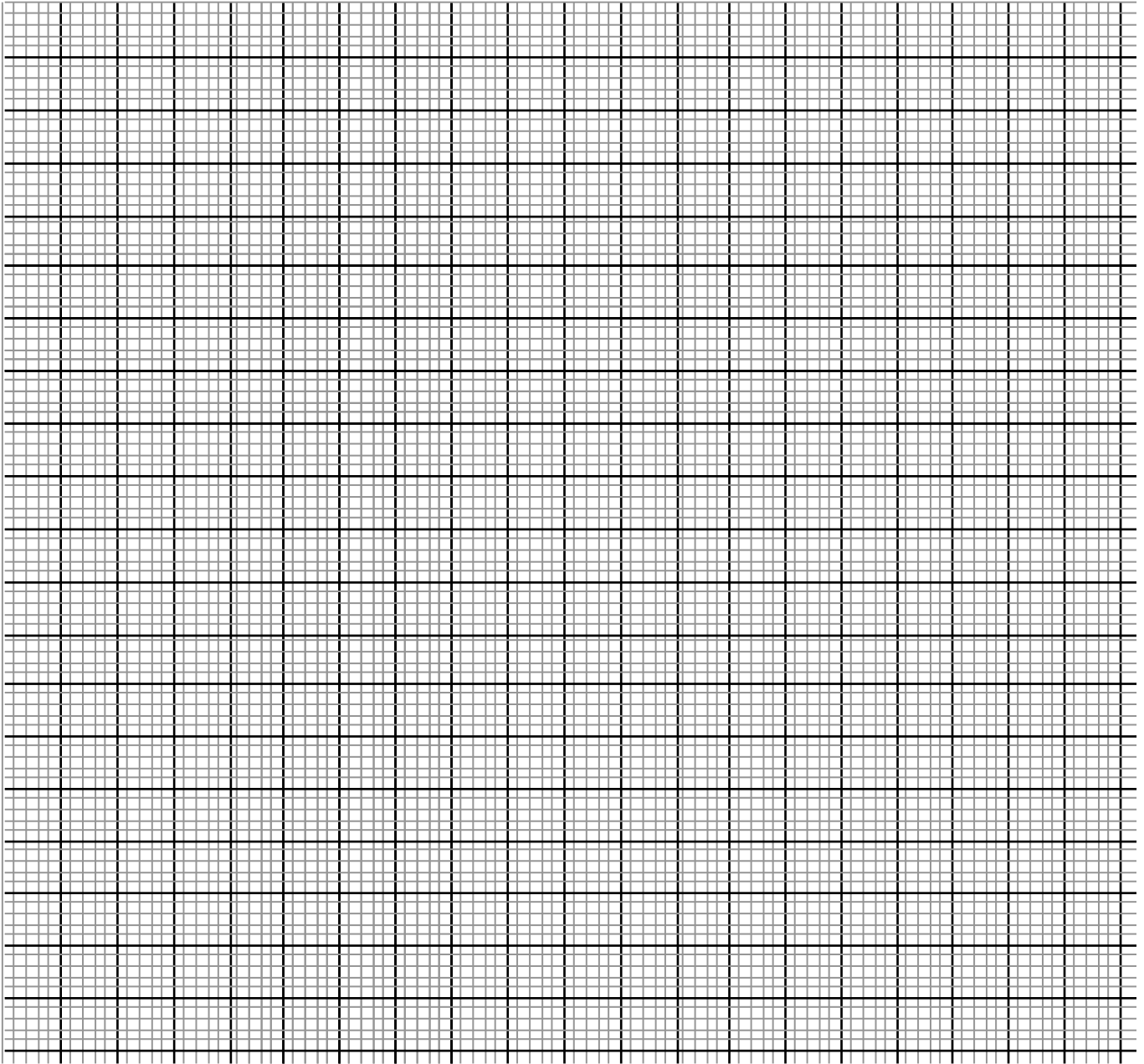
g) Repeat steps **e** and **f** for other values of current as shown on the **table 1** below

Current I (mA)	I₀	$\frac{4}{5}I_0$	$\frac{3}{4}I_0$	$\frac{2}{3}I_0$	$\frac{1}{2}I_0$	$\frac{2}{5}I_0$	$\frac{1}{3}I_0$	$\frac{1}{4}I_0$
Your calculated fraction of I₀ (mA)								
Time t (s)								

(8marks) Table 1

h) Plot a graph of current **I** (y-axis) (**mA**) against time **t** (s)

(5marks)



i) From your graph, find **W** the value of **I** when **t** = **7.00s** in **SI** units. (2marks)

j) Given that **A** = **10W**, determine the value of **A**. (1mark)

k) Determine the voltage across **R** at **t = 7.00s** given that **R = 4.7 KΩ** (2marks)

l) State the quantity represented by the area under the graph (1mark)

2. QUESTION TWO

This question has two parts A and B. Answer both parts.

PART A

You are provided with the following:-

- A retort stand, clamp and boss.
- A spiral spring.
- A stop watch.
- Three 100g masses.

Proceed as follows:

(a) Suspend a 100g mass at the end of the spring as shown in figure 1.

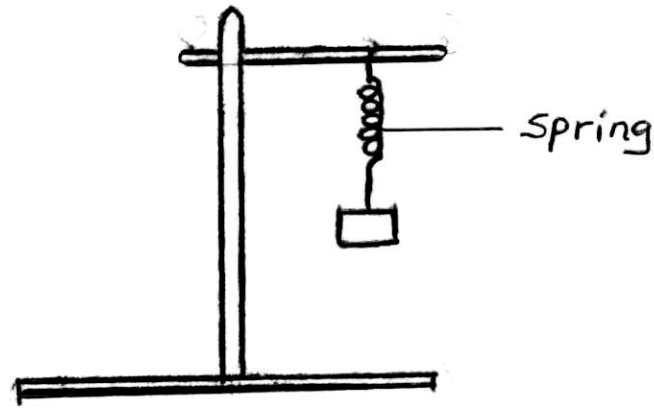


Fig. 1

Now give the mass a small vertical displacement and release so that it performs vertical oscillations. Time ten oscillations and determine the period T . Enter your results in table 1.

(b) Repeat the experiment for the other values of mass and complete the table.

Table 1

Mass m (g)	100	200	300
Time for 10 oscillations (s)			
Periodic time T (s)			

(3mks)

(c) Given that $T = \pi \sqrt{\frac{m}{k}}$, where k is the spring constant, find the average value k for the spring.
K **(2mks)**

PART B

You are provided with the following:-

- A 250ml glass beaker
- A Bunsen burner
- A Thermometer
- A Stop watch
- A Tripod stand and a Wire gauze
- A measuring cylinder 100ml
- Water

Set the apparatus as shown in figure 2 below.

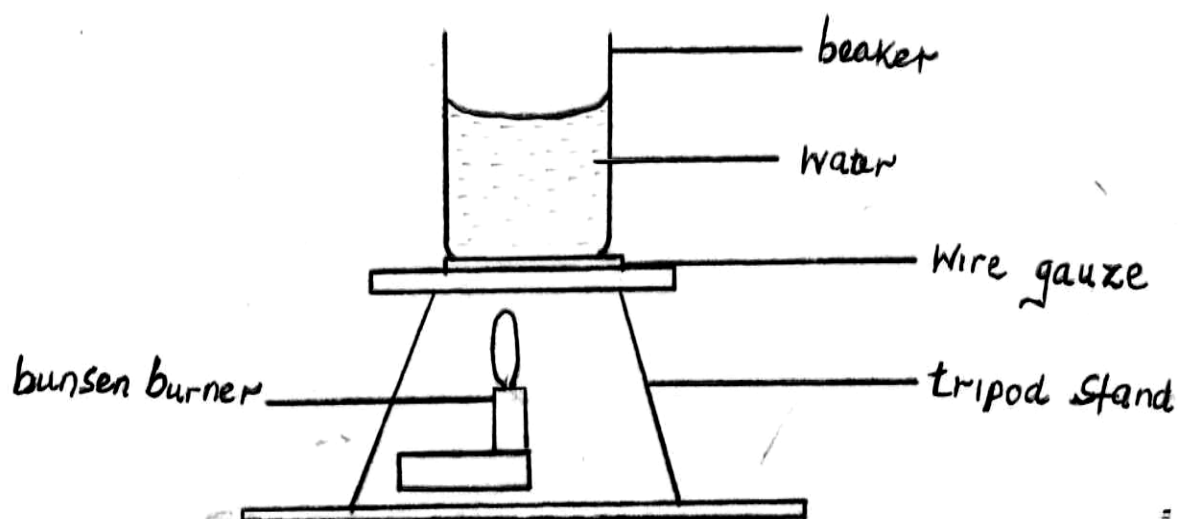


Fig. 2

(d) Measure 100cm^3 of water and pour it into the beaker. Take the initial temperature of the water.

$T_0 = \dots\dots\dots^\circ\text{C}$ (1mk)

Now heat the water to a temperature of 80°C . switch off the gas tap and place a thermometer into the beaker and start the stop watch when the temperature is 65°C . Take the temperature $T^\circ\text{C}$ of water ever two minutes. Record your results in the table 2 below.

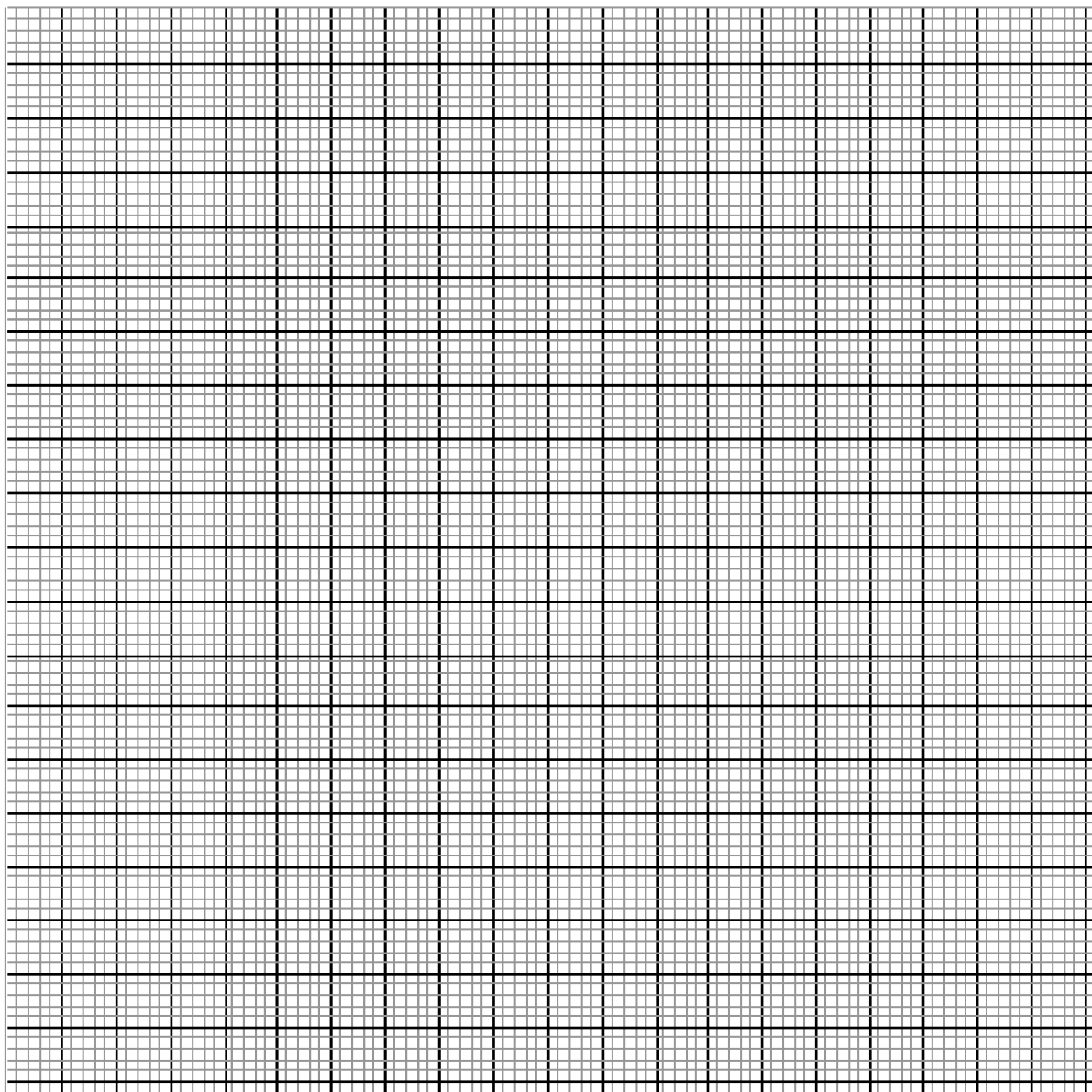
Table 2

Time, t(min)	2	4	6	8	10	12	14
Temperature (T) °C							
(T - T ₀)							
Log (T - T ₀)							

(5mks)

(e) Plot a graph of Log (T - T₀) against Time (t).

(5mks)



(a) Find the value of P of $\log (T - T_0)$ when $t = 0$. **(1mk)**

(b) Determine N , where N is the antilog of P . **(1mk)**

(c) Calculate the temperature of the surrounding T_R using the expression $N = 65 - T_R$ **(2mks)**